

Class Imbalance and Domain Shift in Vehicle-Dynamics-Based Drowsy Driving Detection

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Abstract—Vehicle-dynamics-based drowsy driving detection (DDD) is a non-intrusive and practical approach to driver monitoring, but its performance may be degraded by two issues: class imbalance, because drowsy samples are rare, and domain shift, because driving behavior varies across drivers. In vehicle-dynamics-based DDD, however, it remains unclear whether these issues materially affect detection performance and which countermeasures are effective. To address this question, we construct a DDD framework using a public 87-driver simulator dataset and conduct a four-factor factorial experiment varying distance metric (D), domain membership (G), training mode (M), and rebalancing strategy (R). Sobol–Hoeffding variance decomposition shows that training mode and rebalancing account for more than 95% of systematic variance, with a substantial $R \times M$ interaction that reverses the optimal rebalancing strategy across training modes. In contrast, distance metric and domain membership have no measurable effect on performance. A subject-wise application of SMOTE (SW-SMOTE) at ratio $r = 0.1$ with within-domain training maximizes threshold-independent discrimination (AU-ROC = 0.903, AUPRC = 0.648). These results establish a clear design priority for vehicle-dynamics-based DDD: class rebalancing and training mode selection are the dominant performance levers, whereas domain grouping configuration can be safely omitted from the optimization process.

Index Terms—Drowsy driving detection, class imbalance, domain shift, vehicle dynamics, Sobol sensitivity analysis, factorial experiment, SMOTE.

I. Introduction

DROWSY driving is a leading cause of traffic accidents worldwide, accounting for an estimated 15–20% of road fatalities [1]. Vehicle-dynamics-based drowsy driving detection (DDD) offers a non-intrusive and practical approach by monitoring signals such as steering angle, lateral acceleration, and lane offset without requiring sensors attached to the driver [2], [3].

Two challenges are potentially critical for vehicle-dynamics-based DDD. First, class imbalance: drowsy episodes are inherently rare in driving data, with alert states far outnumbering drowsy states, which may bias classifiers toward the majority class and impair detection of safety-critical events. Second, domain shift: driving behavior varies across individuals. Because training data typically cover only a subset of potential drivers (in-domain), a deployed model must generalize to unseen

drivers (out-domain), whose driving patterns may differ substantially. This inter-driver variability may degrade detection performance.

Improving detection performance under these conditions is a practical priority, yet the degree to which each factor actually degrades performance has not been quantified. Although class imbalance handling [4], [5] and domain adaptation [6], [7] have been studied extensively in other contexts, their importance in vehicle-dynamics-based DDD remains unclear. In particular, it is not known whether these issues materially degrade performance in this modality, whether they warrant dedicated countermeasures, and how they interact when both are present. To this end, we pose the following research questions:

- RQ1. To what extent do class imbalance and domain shift each affect vehicle-dynamics-based DDD performance?
- RQ2. Do the contributing factors interact, and if so, what is the optimal factor combination?

This paper addresses these research questions by constructing a DDD framework using a public open dataset and conducting a four-factor factorial experiment to quantify the effects of class imbalance handling and domain configuration on detection performance. The four factors—distance metric (D), domain membership (G), training mode (M), and rebalancing strategy (R)—are systematically varied across 1,512 evaluated random forest models (126 factor cells $\times$ 12 seeds), and Sobol–Hoeffding variance decomposition is applied to identify which factors significantly affect performance and which combinations are practically effective. Of these, distance metric (D), domain membership (G), and training mode (M) address domain configuration, while rebalancing strategy (R) addresses class imbalance. The main contributions are:

- 1) We construct a DDD experimental framework on a publicly available 87-driver dataset that enables joint, controlled analysis of class imbalance handling and domain shift under a unified factorial design.
- 2) Addressing RQ1, through a four-factor factorial experiment with Sobol–Hoeffding variance decomposition, we quantify the relative importance of these design factors and reveal that training mode and class rebalancing dominate ($> 95\%$ of systematic variance), with a substantial $R \times M$ interaction that reverses the optimal strategy across training modes.
- 3) Addressing RQ2, we position subject-wise SMOTE (SW-SMOTE)—an instance of cluster-based SMOTE [8] with clusters defined by driver identity,

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motivated by the subject-dependency concerns identified for physiological-signal augmentation [9]—within the four-factor design and identify, through exhaustive factor-combination search, SW-SMOTE with within-domain training as the optimal combination, maximizing threshold-independent discrimination (AUROC = 0.903, AUPRC = 0.648).

The remainder of this paper is organized as follows. Section II reviews related work. Section III formalizes the DDD task and the two data challenges (class imbalance and domain shift). Section IV describes the four-factor experimental design, common settings, and analysis methods. Section V reports experimental results from OFAT, Sobol decomposition, interaction analysis, and robustness checks. Section VI answers the research questions and discusses their practical implications. Section VII concludes with a summary and future directions.

II. Related Work

A. Vehicle-Dynamics-Based Drowsy Driving Detection

Vehicle dynamics signals—steering wheel angle, lateral acceleration, and lane position—are widely used for DDD because they enable non-intrusive monitoring. Arefnezhad et al. [10] used statistical and spectral features with SVM, Baccour et al. [11] compared vehicle-based and driver-based features for drowsiness monitoring using SVM with class weighting, and Sun et al. [12] optimized per-driver calculation parameters to account for driver fingerprinting differences. Zhao et al. [13] applied continuous wavelet transform features to steering signals with SVM, and Wang et al. [14] used LSTM on five vehicle signals, although their study targeted driver distraction rather than drowsiness. More recently, Zhu et al. [15] proposed MADFM, a multi-layer adaptive fatigue monitoring model based on steering wheel signals. Recent surveys [2], [3], [16], [17] confirm the practical relevance of this modality. However, most studies evaluate a single method on a fixed split, and the sensitivity of detection performance to data preprocessing choices has received little attention.

B. Class Imbalance in Drowsy Driving Detection

Class imbalance is inherent in drowsy driving data because drowsy episodes are rare but safety-critical. He and Garcia [5] summarize major countermeasures such as random under-sampling (RUS), Synthetic Minority Over-sampling Technique (SMOTE) [4], and cost-sensitive learning, but none of these standard techniques account for the multi-subject structure of pooled driving data. In vehicle-dynamics-based DDD, only a few studies explicitly mention imbalance handling: Yadawadkar et al. [18] compared SMOTE with no resampling, Schwarz et al. [19] used SMOTE in steering-based detection, and Baccour et al. [11] used class weighting when comparing vehicle-dynamics-based and driver-based features. However, these studies treat imbalance handling as an auxiliary choice rather than the main object of study, and systematic comparison of multiple countermeasures remains lacking.

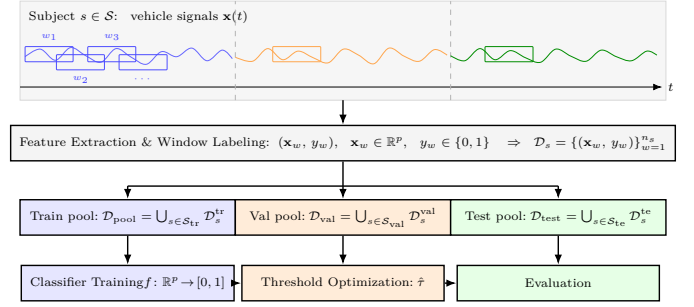


Fig. 1. Generic vehicle-dynamics-based DDD pipeline. Vehicle signals from subject $s \in \mathcal{S}$ are segmented into overlapping sliding windows $w_1, w_2, \dots$; per-window feature vectors $\mathbf{x}_w \in \mathbb{R}^p$ and binary drowsiness labels $y_w \in \{0, 1\}$ are extracted, forming per-subject dataset $\mathcal{D}_s$. A chronological per-subject split pools windows across subjects into training pool $\mathcal{D}_{\text{pool}}$, validation pool $\mathcal{D}_{\text{val}}$, and test pool $\mathcal{D}_{\text{test}}$. The training pool fits the classifier $f: \mathbb{R}^p \rightarrow [0, 1]$; the validation pool tunes the decision threshold $\hat{\tau}$; the test pool is reserved for final evaluation.

C. Domain Shift in Drowsy Driving Detection

Inter-subject variability creates domain shift in DDD [6], [20]. In vehicle-dynamics-based DDD and related driver-state studies, Sun et al. [12] examined subject-independent drowsiness detection while highlighting driver fingerprinting differences, and Maritsch et al. [21] used leave-one-subject-out evaluation for vehicle-dynamics-based driver-state prediction. These studies recognize cross-subject variability, but systematic comparison of domain-related design choices in vehicle-dynamics-based DDD remains limited.

III. Problem Formulation

A. DDD Framework Overview

Vehicle-dynamics-based DDD monitors vehicle signals—such as steering angle, lateral acceleration, and lane position—to infer the driver's alertness level without requiring on-body sensors. Fig. 1 illustrates the generic DDD processing pipeline shared, with minor variations, across most prior vehicle-dynamics-based DDD studies. Vehicle signals from each subject s are segmented into overlapping sliding windows and labeled, yielding per-subject datasets $\mathcal{D}_s$. The datasets are then split chronologically and pooled across subjects into three sets—training pool $\mathcal{D}_{\text{pool}}$, validation pool $\mathcal{D}_{\text{val}}$, and test pool $\mathcal{D}_{\text{test}}$ —which feed classifier training, threshold optimization, and final evaluation, respectively.

Let $\mathcal{S} = \{s_1, \dots, s_N\}$ denote the set of N subjects. For each subject s , vehicle signals are segmented into fixed-length time windows, and each window is assigned a binary drowsiness label based on a subjective sleepiness scale, yielding a per-subject dataset $\mathcal{D}_s = \{(\mathbf{x}_w, y_w)\}_{w=1}^{n_s}$, where $\mathbf{x}_w \in \mathbb{R}^p$ is a p -dimensional feature vector extracted from the window (the value of p and the feature set are specified in Section IV-C2) and $y_w \in \{0, 1\}$ is the drowsiness label (0: alert, 1: drowsy).

Each subject's dataset is split chronologically into three disjoint portions:

$$\mathcal{D}_s = \mathcal{D}_s^{\text{tr}} \dot{\cup} \mathcal{D}_s^{\text{val}} \dot{\cup} \mathcal{D}_s^{\text{te}} \quad (1)$$

with a 70/15/15 ratio, preserving temporal order to prevent data leakage from future windows. The training set $\mathcal{D}_s^{\text{tr}}$ is used to fit the classifier (with hyperparameter selection performed via cross-validation within $\mathcal{D}_s^{\text{tr}}$), the validation set $\mathcal{D}_s^{\text{val}}$ for threshold tuning, and the test set $\mathcal{D}_s^{\text{te}}$ for final performance evaluation.

The DDD task is to learn a scoring function $f: \mathbb{R}^k \rightarrow [0, 1]$ (with $k \leq p$ the input dimensionality after feature selection; Section IV-C2) that generalizes to held-out test windows, trained on pooled data $\mathcal{D}_{\text{pool}} = \bigcup_{s \in \mathcal{S}_{\text{tr}}} \mathcal{D}_s^{\text{tr}}$, where $\mathcal{S}_{\text{tr}} \subseteq \mathcal{S}$ is the training subject set determined by the training-mode factor M (introduced as a study factor in Section IV-B; see Eq. (7)). The binary prediction is obtained by thresholding the calibrated form of $f(\mathbf{x}_w)$: $\hat{y}_w = \mathbb{I}[\hat{p}_w^{\text{cal}} \geq \hat{\tau}]$, where $\mathbb{I}[\cdot]$ is the indicator function, $\hat{p}_w^{\text{cal}}$ is the calibrated posterior probability derived from $f(\mathbf{x}_w)$, and $\hat{\tau}$ is the F2-optimal operating point (the calibration map and the F2 metric are defined in Section IV-C3). The specific dataset, signal set, and preprocessing parameters are detailed in Section IV-C.

B. Class Imbalance

For a given dataset $\mathcal{D}$, let $n_c(\mathcal{D}) = |\{(\mathbf{x}_w, y_w) \in \mathcal{D} : y_w = c\}|$ denote the number of windows with label c , where $c = 0$ (alert) and $c = 1$ (drowsy). Because drowsy episodes are rare, $n_1 \ll n_0$, yielding an imbalance ratio $\rho = n_1/n_0 \ll 1$. A rebalancing operator transforms the training set $\mathcal{D}$ into $\tilde{\mathcal{D}}$ so that the adjusted ratio $\tilde{n}_1/\tilde{n}_0$ approaches a target value r , either by under-sampling the majority class or over-sampling the minority class.

C. Domain Shift

Domain definition. In this paper, we adopt a per-subject notion of domain: each subject $s \in \mathcal{S}$ constitutes one domain, characterized by the marginal feature distribution $P_s(\mathbf{x})$ that the subject's driving behavior induces over the feature space. A subject group such as $\mathcal{G}_{\text{src}} \subseteq \mathcal{S}$ is therefore an aggregation of domains rather than a single domain; the in-domain and out-domain groups introduced in Section IV-B1 likewise denote subject aggregations selected by their distributional characteristics. For brevity we adopt the conventional compound terms “in-/out-domain group” (for $\mathcal{G}_{\text{in}}$, $\mathcal{G}_{\text{out}}$) and “within-/cross-/mixed-domain training” (for the training-mode levels in Section IV-B3); in every case “domain” refers to one of these subject aggregations, never to a single subject's $P_s(\mathbf{x})$.

Inter-driver variability means that $P_{s_i}(\mathbf{x}) \neq P_{s_j}(\mathbf{x})$ for some pairs of subjects (s_i, s_j) ; following the transfer-learning taxonomy of Pan and Yang [6], this constitutes covariate shift under the standard assumption that the labeling mechanism $P(y | \mathbf{x})$ is shared across subjects. When subjects are partitioned into a source group $\mathcal{G}_{\text{src}} \subseteq \mathcal{S}$ and a target group $\mathcal{G}_{\text{tgt}} \subseteq \mathcal{S}$ with $\mathcal{G}_{\text{src}} \cap \mathcal{G}_{\text{tgt}} = \emptyset$, a classifier

trained on $\bigcup_{s \in \mathcal{G}_{\text{src}}} \mathcal{D}_s$ may degrade on $\bigcup_{s \in \mathcal{G}_{\text{tgt}}} \mathcal{D}_s$ because the pooled training distribution differs from the target distribution.

Both class imbalance (Section III-B) and domain shift (Section III-C) are treated here not as fixed nuisances but as explicit design factors whose practical importance is quantified through the factorial experiment in Section IV.

IV. Experimental Framework

To answer the two research questions posed in Section I—which factors materially affect performance (RQ1) and whether their interactions determine the optimal design (RQ2)—we design a four-factor factorial experiment. Sobol–Hoeffding variance decomposition (Section IV-D) quantifies each factor's individual and interactive contribution, enabling a principled answer to both questions from a single unified experiment.

Section IV-A provides an overview of the factorial design; Section IV-B defines each factor; Section IV-C describes common experimental settings; and Section IV-D presents the analysis methods.

A. Overview of the Factorial Experiment

The experiment follows a four-factor full-factorial design. Fig. 2 illustrates the subject partitioning step performed by D , and Fig. 3 shows how G and M jointly determine the training, validation, and test sets for the dependency chain $D \rightarrow G \rightarrow M \rightarrow R$ of the four study-specific factors—distance metric (D), domain membership (G), training mode (M), and rebalancing strategy (R)—where each factor acts on the data partition constructed by its predecessors; the resulting (D, G, M, R) cell is then evaluated using the generic DDD pipeline of Fig. 1. This chain operationalizes the formulation of Section III as follows.

Starting from the subject set $\mathcal{S}$ with per-subject datasets $\mathcal{D}_s = \{(\mathbf{x}_w, y_w)\}$ (Section III-A), a distance metric D first computes pairwise inter-subject distances and partitions $\mathcal{S}$ into an in-domain group $\mathcal{G}_{\text{in}}$ (44 subjects, most typical) and an out-domain group $\mathcal{G}_{\text{out}}$ (43 subjects, most dissimilar). Domain membership G then selects which of these two groups serves as the evaluation target $\mathcal{G}_{\text{eval}}$. Each subject's data is split chronologically into $(\mathcal{D}_s^{\text{tr}}, \mathcal{D}_s^{\text{val}}, \mathcal{D}_s^{\text{te}})$ per Eq. (1) (70/15/15 ratio). Training mode M assembles the pooled training set $\mathcal{D}_{\text{pool}} = \bigcup_{s \in \mathcal{S}_{\text{tr}}} \mathcal{D}_s^{\text{tr}}$ from the subject set $\mathcal{S}_{\text{tr}}$ defined jointly by G and M (Eq. (7)). After feature extraction and selection (from $p = 145$ to $k = 10$ features per Section IV-C2), rebalancing strategy R applies the operator of Section III-B to produce $\tilde{\mathcal{D}}$ with adjusted class ratio approaching the target r . The scoring function $f: \mathbb{R}^k \rightarrow [0, 1]$ (Section III-A) is trained on $\tilde{\mathcal{D}}$, calibrated, and evaluated on $\mathcal{D}_{\text{test}} = \bigcup_{s \in \mathcal{G}_{\text{eval}}} \mathcal{D}_s^{\text{te}}$.

As shown in Fig. 3, classifier training depends on (G, M, R) through the composition of $\tilde{\mathcal{D}}$, threshold optimization depends on (G, M) through the validation pool, and evaluation depends only on G through $\mathcal{G}_{\text{eval}}$. Each factorial cell is uniquely determined by a quadruple

TABLE I

Four-Factor Full-Factorial Experimental Design ($3 \times 2 \times 3 \times 7 = 126$ cells; 12 fixed seeds; $N = 1,512$ total observations). Sym.: factor symbol; Lvl.: number of levels.

Factor	Sym.	Lvl.	Values
Distance	D	3	MMD, DTW, Wasserstein-1 (W_1)
Membership	G	2	In-domain, Out-domain
Training mode	M	3	Cross, Within, Mixed
Rebalancing	R	7	BL, RUS/SMOTE/SW-SMOTE $r \in \{0.1, 0.5\}$

(D, G, M, R); Section IV-B defines the levels of each factor (summarized in Table I) and the total number of observations.

B. Factor Definitions

Table I summarizes the four factors and their levels. The $3 \times 2 \times 3 \times 7 = 126$ factor combinations are each evaluated over 12 fixed random seeds, yielding $N = 1,512$ total observations. The following subsubsections define each factor in detail.

1) Distance Metric (D) and Domain Grouping: To define the binary domain partition ($\mathcal{G}_{\text{in}}, \mathcal{G}_{\text{out}}$) on which factors G and M operate, pairwise distances $d(s_i, s_j)$ between all 87 subjects are computed from the per-subject window sets $\mathcal{D}_{s_i}$ and $\mathcal{D}_{s_j}$ using three metrics:

Maximum Mean Discrepancy (MMD) [22] measures distributional difference in a reproducing kernel Hilbert space $\mathcal{H}$:

$$\text{MMD}^2(s_i, s_j) = \left\| \frac{1}{n_{s_i}} \sum_{(\mathbf{x}_w, \mathbf{y}_w) \in \mathcal{D}_{s_i}} \phi(\mathbf{x}_w) - \frac{1}{n_{s_j}} \sum_{(\mathbf{x}_w, \mathbf{y}_w) \in \mathcal{D}_{s_j}} \phi(\mathbf{x}_w) \right\|_{\mathcal{H}}^2 \quad (2)$$

where ϕ is the feature map induced by a Gaussian kernel.

Dynamic Time Warping (DTW) [23] aligns two univariate time series and finds the warping path that minimizes the root sum-of-squared distances along the optimal alignment. Each subject's multivariate window sequence (indexed $t = 1, \dots, n_{s_i}$ over $\mathcal{D}_{s_i}$) is first reduced to a scalar series $\bar{x}_t^{(i)} = \frac{1}{q} \sum_{d=1}^q x_{t,d}^{(i)}$ (mean across all $q = 145$ z -scored features at each time step), and DTW is computed as:

$$\text{DTW}(s_i, s_j) = \sqrt{\min_{\pi \in \mathcal{P}} \sum_{(a,b) \in \pi} (\bar{x}_a^{(i)} - \bar{x}_b^{(j)})^2} \quad (3)$$

where $\mathcal{P}$ is the set of admissible warping paths. Because DTW with Euclidean cost is symmetric, no explicit symmetrization is required.

Wasserstein Distance [24] computes the minimum transport cost between the empirical feature distributions:

$$W_1(s_i, s_j) = \inf_{\gamma \in \Gamma(P_{s_i}, P_{s_j})} \int \|\mathbf{x} - \mathbf{x}'\| d\gamma(\mathbf{x}, \mathbf{x}') \quad (4)$$

where $\Gamma(P_{s_i}, P_{s_j})$ is the set of couplings with marginals P_{s_i} and P_{s_j} .

For each metric, a K -nearest-neighbor (KNN) score ($K = 5$) is computed per subject as the mean distance to its K nearest neighbors:

$$\text{KNN}_K(s) = \frac{1}{K} \sum_{j \in \mathcal{N}_K(s)} d(s, s_j) \quad (5)$$

Subjects are split at the median of KNN_K into in-domain (44 subjects, lowest scores, most typical) and out-domain (43 subjects, highest scores, most dissimilar), forming two groups $\mathcal{G}_{\text{in}}$ (in-domain, 44 subjects) and $\mathcal{G}_{\text{out}}$ (out-domain, 43 subjects); their roles as evaluation target or training source are determined subsequently by G and M (Eqs. (6)–(7)).

Distance metric $D \in \{\text{MMD}, \text{DTW}, W_1\}$ determines the pairwise subject distances and the binary typicality-based partition ($\mathcal{G}_{\text{in}}, \mathcal{G}_{\text{out}}$) via Eqs. (2)–(5).

2) Domain Membership (G): Domain membership $G \in \{\text{in}, \text{out}\}$ selects the evaluation-target group:

$$\mathcal{G}_{\text{eval}} = \begin{cases} \mathcal{G}_{\text{in}} & \text{if } G = \text{in} \\ \mathcal{G}_{\text{out}} & \text{if } G = \text{out} \end{cases} \quad (6)$$

where $|\mathcal{G}_{\text{in}}| = 44$ and $|\mathcal{G}_{\text{out}}| = 43$ partition $\mathcal{S}$ ($|\mathcal{S}| = 87$; $\mathcal{G}_{\text{in}} \cap \mathcal{G}_{\text{out}} = \emptyset$, $\mathcal{G}_{\text{in}} \cup \mathcal{G}_{\text{out}} = \mathcal{S}$).

3) Training Mode (M): Training mode $M \in \{\text{Cross}, \text{Within}, \text{Mixed}\}$ assembles the training pool from subjects determined by $\mathcal{G}_{\text{eval}}$:

$$\mathcal{D}_{\text{pool}} = \bigcup_{s \in \mathcal{S}_{\text{tr}}} \mathcal{D}_s^{\text{tr}}, \quad \mathcal{S}_{\text{tr}} = \begin{cases} \mathcal{S} \setminus \mathcal{G}_{\text{eval}} & M = \text{Cross} \\ \mathcal{G}_{\text{eval}} & M = \text{Within} \\ \mathcal{S} & M = \text{Mixed} \end{cases} \quad (7)$$

Cross-domain trains exclusively on the opposite group, in-domain on the evaluation group itself, and mixed on all 87 subjects. The subject overlap ratio

$$\omega_{\text{tr}} = \frac{|\mathcal{S}_{\text{tr}} \cap \mathcal{G}_{\text{eval}}|}{|\mathcal{G}_{\text{eval}}|} \quad (8)$$

quantifies subject leakage between training and evaluation: $\omega_{\text{tr}} = 0$ for $M = \text{Cross}$, and $\omega_{\text{tr}} = 1$ for $M \in \{\text{Within}, \text{Mixed}\}$.

4) Rebalancing Strategy (R): Rebalancing strategy R applies a rebalancing operator $\mathcal{R}$ to the training pool, yielding $\tilde{\mathcal{D}} = \mathcal{R}(\mathcal{D}_{\text{pool}})$ with adjusted class ratio $\tilde{n}_1/\tilde{n}_0$. Seven levels are defined:

- Baseline (BL): $\tilde{\mathcal{D}} = \mathcal{D}_{\text{pool}}$ (identity; original ratio ρ preserved).
- RUS at $r \in \{0.1, 0.5\}$: randomly discard majority-class samples until $\tilde{n}_1/\tilde{n}_0 = r$.
- SMOTE [4] at $r \in \{0.1, 0.5\}$: generate synthetic minority-class samples via k -NN interpolation on $\mathcal{D}_{\text{pool}}$ until $\tilde{n}_1/\tilde{n}_0 = r$.
- SW-SMOTE at $r \in \{0.1, 0.5\}$: apply SMOTE independently within each subject before pooling, an instance of cluster-based SMOTE [8] with clusters fixed to subject identity:

$$\tilde{\mathcal{D}} = \bigcup_{s \in \mathcal{S}_{\text{tr}}} \text{SMOTE}_r(\mathcal{D}_s^{\text{tr}}) \quad (9)$$

TABLE II
Five Vehicle-Dynamics Signals Used as Classifier Inputs (sampled at $f_s = 60$ Hz)

Symbol	Signal	Unit
$\delta(t)$	Steering angle	rad
$\dot{\delta}(t)$	Steering speed	rad/s
$a_y(t)$	Lateral acceleration	m/s ²
$a_x(t)$	Longitudinal acceleration	m/s ²
$e_{\text{lane}}(t)$	Lane offset	m

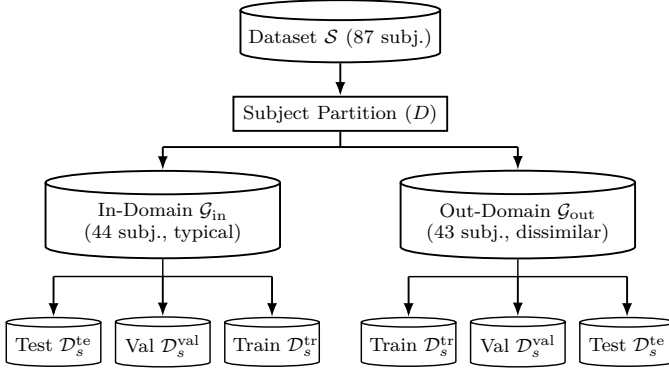


Fig. 2. Subject partitioning by factor D : the dataset $\mathcal{S}$ (87 subjects) is split by the median KNN score under distance metric D into $\mathcal{G}_{\text{in}}$ (44 subjects, most typical) and $\mathcal{G}_{\text{out}}$ (43 subjects, most dissimilar); each subject's data is then chronologically partitioned into Train, Val, and Test at a 70/15/15 ratio.

so that synthetic samples preserve individual driving patterns rather than interpolating across subjects.

The classifier f is trained on $\tilde{\mathcal{D}}$ and evaluated on $\mathcal{D}_{\text{test}} = \bigcup_{s \in \mathcal{G}_{\text{eval}}} \mathcal{D}_s^{\text{te}}$. The validation and test sets are kept in their original class distribution throughout; applying any resampling operator to evaluation data would misrepresent the deployment-time imbalance ratio and yield optimistic estimates of recall-oriented metrics [4], [5].

C. Common Experimental Settings

1) **Dataset and Signals:** The publicly available driving dataset of Aygun et al. [25] is used, comprising $N = 87$ subjects each recorded over two sessions on a medium-fidelity partial-cab driving simulator at $f_s = 60$ Hz. Five raw signals are extracted (Table II).

Under the kinematic bicycle model [26], lateral acceleration follows:

$$a_y = \frac{v^2}{L} \delta \quad (10)$$

where v is vehicle speed, L is the wheelbase, and the small-angle approximation $\tan \delta \approx \delta$ is applied ($|\delta| \ll 1$). Lane offset evolves as:

$$\dot{e}_{\text{lane}}(t) = a_y(t) - a_{y,\text{road}}(t) \quad (11)$$

Because $\dot{\delta} = d\delta/dt$ and $a_y \propto \delta$, four of the five signals ($\delta, \dot{\delta}, a_y, e_{\text{lane}}$) are deterministically coupled through a single degree of freedom $\delta(t)$, with only a_x dynamically independent.

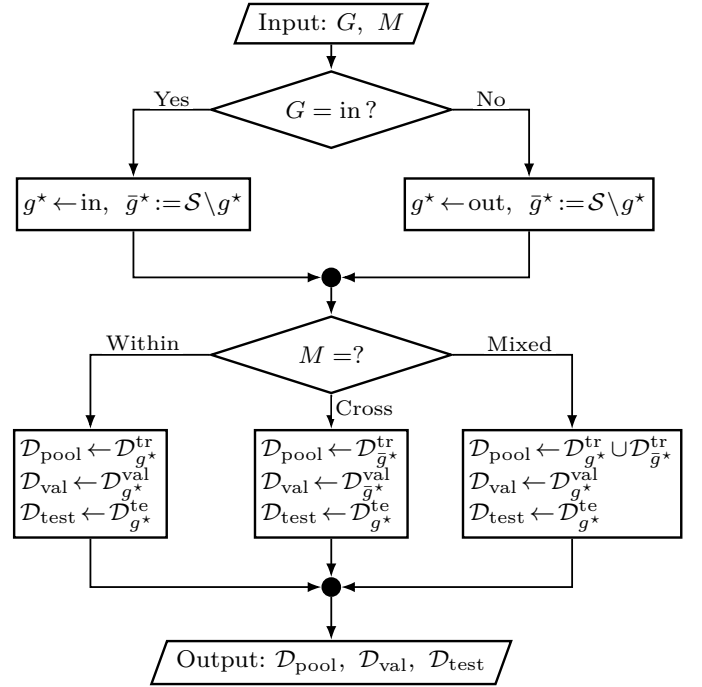


Fig. 3. Training and evaluation set assembly determined by factors G and M (ISO 5807 flowchart). $G \in \{\text{in}, \text{out}\}$ selects the evaluation-target group g^* and its complement $\bar{g}^* = \{\text{in}, \text{out}\} \setminus \{g^*\}$; $\mathcal{D}_g^{\text{tr}}, \mathcal{D}_g^{\text{val}}, \mathcal{D}_g^{\text{te}}$ denote the Train/Val/Test partitions of group g . Factor M then determines the composition of the training pool: Within uses the target group's train split; Cross uses the complement group's train and val splits, with test from the target group; Mixed uses the union of both groups' train splits with val and test from the target group (Eq. (7)).

2) **Data Preprocessing:** Raw signals are segmented into 3-second sliding windows with 50% overlap (1.5-second step). Each window is labeled using the Karolinska Sleepiness Scale (KSS) [27]: $\text{KSS} \in \{1, \dots, 5\}$ maps to Alert (class 0) and $\text{KSS} \in \{8, 9\}$ to Drowsy (class 1); intermediate scores (6, 7) are excluded to maximize label reliability.

For each window, 18 statistical and spectral descriptors (following the feature-extraction methodology of Arefnezhad et al. [10], extended here from the original two steering signals to all five vehicle signals) are extracted from each of the five vehicle signals ($\delta, \dot{\delta}, a_y, a_x, e_{\text{lane}}$), yielding $5 \times 18 = 90$ statistical and spectral descriptors. Each signal's raw waveform and its FFT power spectrum in the $[0.5, 30]$ Hz band produce 18 descriptors: Range, StdDev, Energy, Zero-Crossing Rate, Quartiles (Q25, Median, Q75), Skewness, Kurtosis, Katz Fractal Dimension, Shannon Entropy, Frequency Variance, Spectral Entropy, Spectral Flux, Frequency Center of Gravity, Dominant Frequency, Average PSD, and Sample Entropy. Additionally, $5 \times 3 = 15$ smoothed-trend / standard-deviation / prediction-error descriptors and $5 \times 8 = 40$ Geronimo-Hardin-Massopust (GHM) multiwavelet packet decomposition energies (3-level decomposition, 8 sub-bands per signal) are extracted from the same five signals, giving a unified pool of $p = 90 + 15 + 40 = 145$ features that serves both as the RF classifier input and as the

inter-subject distance-computation space (Section IV-B1). To reduce dimensionality and mitigate overfitting, the $p = 145$ features are ranked by random forest feature importance [28] (200 trees, balanced class weights) and the top $k = 10$ are retained for each training configuration, balancing discriminative power against overfitting given the per-cell sample sizes.

3) Classifier and Hyperparameter Optimization: The scoring function $f: \mathbb{R}^k \rightarrow [0, 1]$ ($k = 10$ retained features; Section IV-C2) introduced in Section III is instantiated as a random forest (RF) [28] and trained on the rebalanced pool $\tilde{\mathcal{D}}$ defined above. Its hyperparameter vector θ is tuned independently per factorial cell via Optuna [29] with 100 Tree-structured Parzen Estimator (TPE) trials and 3-fold stratified cross-validation:

$$\hat{\theta} = \arg \max_{\theta \in \Theta} \text{F2}_{\text{CV}}(\theta; \tilde{\mathcal{D}}) \quad (12)$$

where Θ is the search space defined in Table III. After Optuna selection and Platt-scaling calibration (described below), the classification threshold τ is swept over the set of unique calibrated posterior probabilities derived from the precision–recall curve on the validation set, retaining the value that maximizes F2:

$$\hat{\tau} = \arg \max_{\tau \in [0, 1]} \text{F2}\left(\{(\hat{p}_w^{\text{cal}}, y_w) : (\mathbf{x}_w, y_w) \in \bigcup_{s \in \mathcal{G}_{\text{eval}}} \mathcal{D}_s^{\text{val}}\}, \tau\right) \quad (13)$$

where $\hat{p}_w^{\text{cal}}$ is the calibrated posterior probability (Platt scaling, described below). Calibration precedes threshold optimization: the base RF is calibrated via Platt scaling [30] using CalibratedClassifierCV with a fresh 5-fold stratified partition (random_state=42) of the train+validation pool, independent of the 3-fold HPO partition—which uses train only, so the validation split never enters HPO. Within each calibration fold, the base classifier is refit on 4/5 of the combined data and a sigmoid is fit on the held-out 1/5; the final calibrated probability $\hat{p}_w^{\text{cal}}$ is the mean across the five fold-specific calibrators. During HPO, Optuna co-explores class_weight $\in \{\text{balanced}, \text{balanced_subsample}\}$ jointly with the seven other hyperparameters in Table III, so that the CV objective $\text{F2}_{\text{CV}}(\theta; \tilde{\mathcal{D}})$ is evaluated under standard imbalance-aware tree construction. For the final evaluation, the seven non-class_weight hyperparameters from $\hat{\theta}$ are retained, but class_weight is overridden to a fixed $\{0:1.0, 1:3.0\}$ across all factorial cells. This override deliberately discards the HPO-selected class_weight so that minority-recall emphasis is uniform across the factorial design, decoupling the class-weight axis from the rebalancing factor R (otherwise an Optuna selection of, e.g., balanced versus balanced_subsample per factorial cell would itself become a confounding factor). Each observation in the factorial experiment is therefore:

$$Y_{D,G,M,R} = \text{Metric}\left(\{(\hat{y}_w, y_w) : (\mathbf{x}_w, y_w) \in \mathcal{D}_{\text{test}}\}\right), \quad \hat{y}_w = \mathbb{I}[\hat{p}_w^{\text{cal}} \geq \hat{\tau}] \quad (14)$$

TABLE III
Random Forest Hyperparameter Search Space (Optuna, 100 TPE Trials)

Parameter	Range	Type
n_estimators	100–1,500	int
max_depth	{5, 10, 20, 30, 50, 100, None}	cat
min_samples_split	10–100	int
min_samples_leaf	5–50	int
max_features	$\{\sqrt{k}, \log_2 k, 0.05, 0.1, 0.2, 0.3, 0.5\}$	cat
min_wt_frac_leaf	0.0–0.1	float
max_samples	{None, 0.5, 0.7, 0.9}	cat
class_weight [†]	{balanced, balanced_subsample}	cat
[†] Searched during HPO but overridden to $\{0:1.0, 1:3.0\}$ in the final evaluation (Section IV); the selected value is discarded and only the other seven parameters are carried forward.		

where $\mathcal{D}_{\text{test}} = \bigcup_{s \in \mathcal{G}_{\text{eval}}} \mathcal{D}_s^{\text{te}}$ and Metric $\in \{\text{F2}, \text{AUROC}, \text{AUPRC}\}$.

Computational cost. The factorial design comprises $N = 1,512$ observations ($126 \text{ cells} \times 12 \text{ seeds}$); producing each observation entails 100 Optuna trials with 3-fold CV plus calibration and a final fit, so the entire experiment performs approximately 4.5×10^5 individual RF fits. Individual HPC jobs required 4 CPU cores, 8–10 GB RAM, and 6–8 hours wall time; the entire experiment consumed $\approx 10,000$ core-hours on the university cluster.

Inference latency. A trained RF with $\leq 1,500$ trees and $k = 10$ input features produces a prediction in < 1 ms on a single CPU core, well within the 3-second window update interval.

4) Evaluation Metrics: The observation $Y_{D,G,M,R}$ in Eq. (14) is computed with three complementary metrics chosen to capture different aspects of imbalanced classification:

F2-score is the threshold-dependent, recall-weighted harmonic mean of precision (P) and recall (Rec):

$$\text{F2} = \frac{5 P \text{Rec}}{4 P + \text{Rec}} \quad (15)$$

The $\beta = 2$ weighting penalizes missed drowsy episodes more heavily than false alarms, reflecting the safety-critical nature of DDD. F2 is used as the primary optimization target in Eqs. (12) and (13).

AUROC (area under the ROC curve) summarizes threshold-free discrimination ability over the full operating range. It complements F2 by assessing whether the model ranks drowsy windows above alert ones regardless of threshold choice.

AUPRC (area under the precision–recall curve) [31] provides a threshold-free measure that is more informative than AUROC under severe class imbalance, because it does not credit correct majority-class predictions.

Together, these three metrics form a minimal sufficient set: one threshold-dependent recall-focused metric (F2), one threshold-free global metric (AUROC), and one threshold-free imbalance-aware metric (AUPRC).

D. Analysis Methods

Three analysis methods are applied to the $N = 1,512$ observations: an intuitive OFAT scan, a Sobol–Hoeffding

variance decomposition, and a statistical framework for effect sizes and null-claim support.

1) One-Factor-at-a-Time (OFAT) Analysis: Before the quantitative decomposition, we perform a one-factor-at-a-time (OFAT) analysis [32] to build intuition. For each target factor, all other factors are fixed at every possible combination, and the mean performance is plotted across the target factor's levels. Each resulting line represents one fixed condition (averaged over 12 seeds), producing a “spaghetti plot” that reveals: (i) whether the factor's effect is consistent across conditions, (ii) how much the estimate varies depending on which conditions are fixed, and (iii) the magnitude of the factor's effect relative to the spread across conditions.

2) Variance-Based Sensitivity Analysis (Sobol Indices): To quantify the relative importance of each factor and their interactions, we adopt the Sobol–Hoeffding functional ANOVA decomposition [32], [33]. Let the response surface $Y = g(X_1, \dots, X_m) + \varepsilon$ map factor configurations to performance, where $Y \equiv Y_{D,G,M,R}$ (Eq. (14)) is computed separately for each Metric $\in \{F2, \text{AUROC}, \text{AUPRC}\}$, $m = 4$, $(X_1, X_2, X_3, X_4) = (D, G, M, R)$, and ε is the seed-induced residual variation that contributes the term S_ε in Section V-B. The total variance admits a unique orthogonal decomposition:

$$V(Y) = \sum_{i=1}^m V_i + \sum_{i < j} V_{ij} + \dots + V_{1,2,\dots,m} \quad (16)$$

where $V_i = \text{Var}_{X_i}[\mathbb{E}_{X_{\sim i}}(Y \mid X_i)]$ is the variance of the conditional expectation over factor i alone, and V_{ij} captures the joint effect of (X_i, X_j) not explained by their individual contributions. The first-order Sobol index normalizes each contribution:

$$S_i = \frac{V_i}{V(Y)} \quad (17)$$

and the total-order index aggregates the main effect with all interactions involving factor i :

$$S_{Ti} = 1 - \frac{V_{\sim i}}{V(Y)} \quad (18)$$

where $V_{\sim i} = \text{Var}_{X_{\sim i}}[\mathbb{E}_{X_i}(Y \mid X_{\sim i})]$ is the variance explained by all factors except i . The difference $S_{Ti} - S_i$ quantifies the fraction of variance attributable to interactions involving factor i . Because our design is balanced, all sum-of-squares terms are computed exactly via marginal means without Monte Carlo sampling. Confidence intervals (95%) are obtained by percentile bootstrap ($B = 2,000$) resampling over the 12 seeds.

3) Statistical Framework: Effect sizes between factor levels are measured by Cliff's δ [34] (< 0.147 negligible, < 0.33 small, < 0.474 medium, ≥ 0.474 large). Claims of factor irrelevance (e.g., distance metric equivalence) are supplemented with Bayes factors via BIC approximation [35], interpreted on the Jeffreys scale [36].

Global validation. A permutation test ($B = 10,000$) validates the overall rebalancing effect by shuffling the R labels across observations and recomputing S_R to derive

the null distribution. Seed convergence is assessed by σ_{rank} , the standard deviation of each strategy's rank position across random k -seed subsets averaged over all $\binom{12}{k}$ draws; $\sigma_{\text{rank}} \rightarrow 0$ indicates that rankings are determined by strategy identity rather than seed selection.

Per-feature inter-subject identifiability. For each of the $p = 145$ features in the unified feature pool of Section IV-C2 (90 statistical and spectral descriptors + 15 smooth/std/pred-error + 40 wavelet packet sub-bands), we report a per-feature one-way intraclass correlation, $\text{ICC}_j = \hat{\sigma}_{\text{subject},j}^2 / (\hat{\sigma}_{\text{subject},j}^2 + \hat{\sigma}_{\text{window}|\text{subject},j}^2)$, where $\hat{\sigma}_{\text{subject},j}^2$ is estimated as the variance across the 87 subject-means of feature j and $\hat{\sigma}_{\text{window}|\text{subject},j}^2$ as the average within-subject variance of the same feature (population variance-component form of $\text{ICC}(1,1)$; equivalent to the Shrout–Fleiss mean-square estimator [37] in the large- k limit). The reported overall value $\text{ICC} = \frac{1}{145} \sum_{j=1}^{145} \text{ICC}_j$ summarizes how distinguishable subjects are in the feature space used for inter-subject distance computation; this quantity is referenced once in Section VI-B.

V. Experimental Results

A. Intuitive Factor Analysis via OFAT

Before the quantitative variance decomposition, we screen each factor's effect using OFAT spaghetti plots (Fig. 4). In each panel, every thin line represents one fixed combination of the other three factors (averaged over 12 seeds); the bold line shows the grand OFAT mean and the gray band indicates ± 1 SD. Columns correspond to the four experimental factors; rows show F2-score, AUROC, and AUPRC.

The two left columns show the dominant factors. Rebalancing (R): SMOTE and SW-SMOTE variants consistently outperform Baseline and RUS, but the inter-condition spread ($\Delta = 0.08$ – 0.55 for F2) reveals strong mode-dependence—cross-domain conditions cluster near floor performance while within/mixed conditions disperse upward across a wide range, suggesting an $R \times M$ interaction. Mode (M): every condition rises monotonically from Cross to Within/Mixed, with markedly increased between-condition dispersion at Within/Mixed that mirrors the rebalancing effect.

The two right columns show the negligible factors. Distance (D): the OFAT mean is flat across MMD, DTW, and Wasserstein, and condition lines remain nearly parallel. Membership (G): the mean is near-flat, and individual lines diverge in both directions, averaging to near zero.

B. Quantitative Factor Decomposition via Sobol Indices

The Sobol–Hoeffding decomposition (Fig. 5) quantifies each factor's contribution to total variance. The permutation test confirms a significant global rebalancing effect for all three metrics ($p < 0.001$).

Mode contributes the largest main effect ($S_M = 0.31$ – 0.51), followed by Rebalancing ($S_R = 0.24$ – 0.29). Both factors participate in a substantial $R \times M$ interaction (21.2% of F2-score variance, 13.8% of AUROC variance).

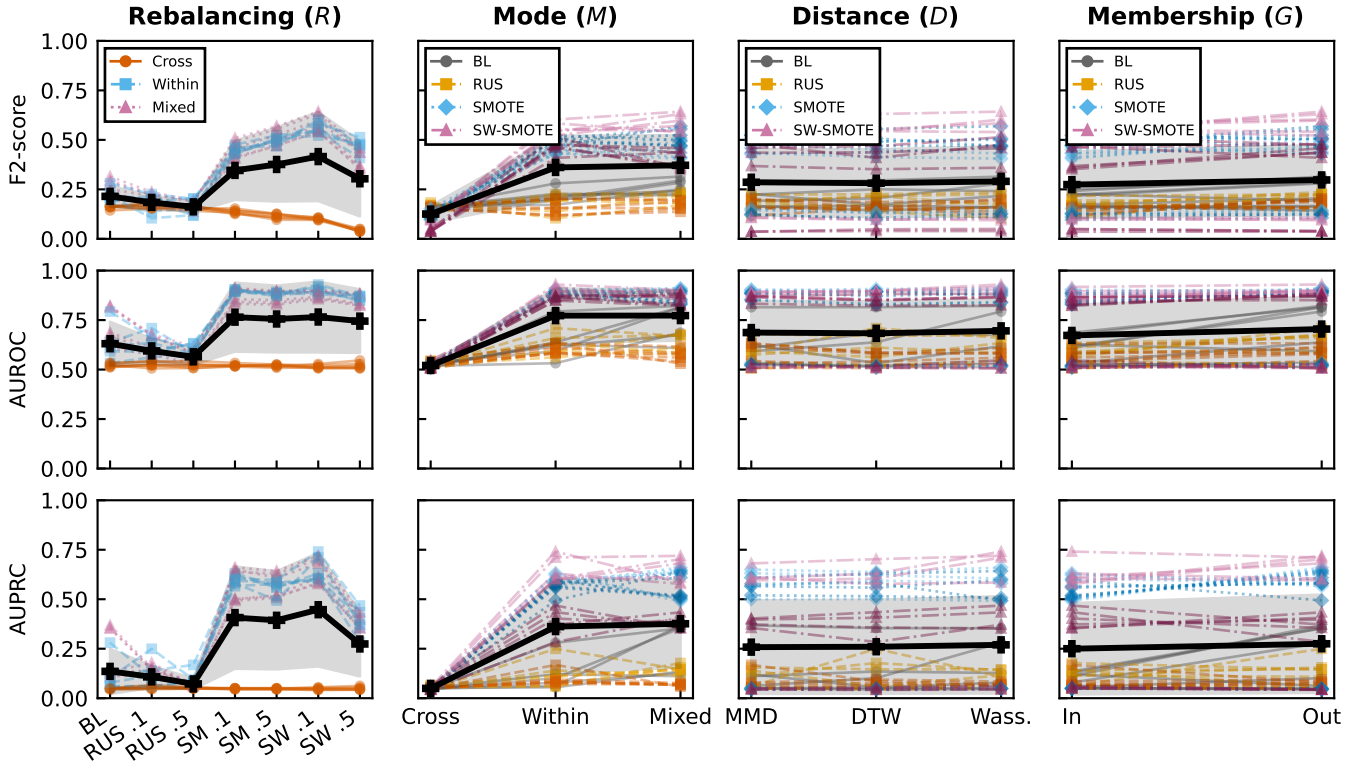


Fig. 4. OFAT spaghetti plots for all four experimental factors (columns: R , M , D , G) and three metrics (rows: F2-score, AUROC, AUPRC). Each thin line represents one fixed combination of the other three factors averaged over 12 seeds. Column R encodes lines by mode (Cross/Within/Mixed) to expose the $R \times M$ interaction; all other columns encode lines by rebalancing family: BL (grey, solid, circle), RUS (orange, dashed, square), SMOTE (sky blue, dotted, diamond), SW-SMOTE (purple, dash-dotted, triangle); ratio variants share colour tone within each family. The bold black line (plus marker) shows the OFAT grand mean; the grey band indicates ± 1 SD. Left two columns (R , M) show dominant factors with large spread; right two columns (D , G) show near-flat trajectories confirming negligible main effects.

The total-order indices show that Mode accounts for $S_{TM} = 0.48\text{--}0.66$ and Rebalancing for $S_{TR} = 0.39\text{--}0.46$ of total variance.

In contrast, Distance ($S_{TD} < 0.015$) and Membership ($S_{TG} \leq 0.031$) are negligible even when interactions are included. Residual variance (seed-to-seed variation), denoted S_ε , accounts for 7.9%–21.9% of total variance.

Defining the systematic (non-residual) variance fraction:

$$\frac{S_M + S_R + S_{R \times M}}{1 - S_\varepsilon} = \frac{0.8224}{0.8433} = 97.5\% \text{ (F2)} \quad (19)$$

where $S_M=0.3675$, $S_R=0.2425$, $S_{R \times M}=0.2124$, and $S_\varepsilon=0.1567$. Analogous ratios are 95.8% (AUROC) and 96.1% (AUPRC), confirming that M , R , and their interaction account for $> 95\%$ of systematic variance across all three metrics. The interaction concentration ratio $C_{R \times M}^{(i)} = S_{R \times M} / (S_{Ti} - S_i)$ for $i \in \{M, R\}$ ranges from 87% to 96%, indicating that the $R \times M$ pair absorbs nearly all interaction variance involving either dominant factor.

Bayes factors further confirm distance metric equivalence, providing “very strong” to “extreme” evidence for H_0 on the Jeffreys scale [36] ($BF_{01} = 71\text{--}767$).

C. $R \times M$ Interaction Analysis

The Sobol decomposition identified a single dominant interaction ($R \times M$). Table IV and Fig. 6 characterize its

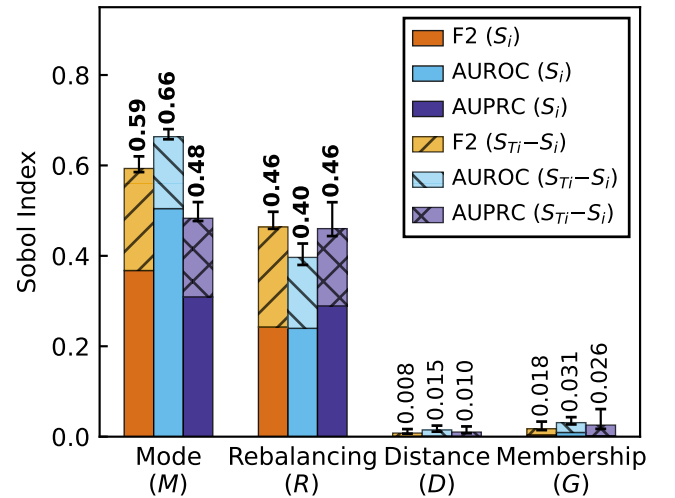


Fig. 5. Sobol sensitivity indices for the four factors. Solid bars: first-order indices (S_i); hatched bars: interaction contributions ($S_{Ti} - S_i$). Error bars: 95% bootstrap CIs on S_{Ti} . Mode and Rebalancing dominate; Distance and Membership are negligible.

structure.

Within-domain and Mixed are statistically equivalent in the pooled comparison ($|\delta_{F2}| = 0.04$, $|\delta_{AUROC}| = 0.05$,

TABLE IV

Mean Performance by Rebalancing Strategy, Disaggregated by Training Mode. Cross-domain results (all $F2 < 0.17$) are shown for completeness; the $R \times M$ interaction analysis focuses on Within and Mixed modes where strategy differentiation is meaningful.

Strategy	F2 (Cross)	F2-score		AUROC		AUPRC	
		Within	Mixed	Within	Mixed	Within	Mixed
Baseline	0.160	0.215	0.268	0.633	0.737	0.116	0.241
RUS $r=0.1$	0.165	0.178	0.208	0.625	0.632	0.122	0.147
RUS $r=0.5$	0.156	0.159	0.170	0.603	0.568	0.097	0.068
SMOTE $r=0.1$	0.138	0.448	0.452	0.898	0.875	0.600	0.573
SMOTE $r=0.5$	0.114	0.499	0.514	0.882	0.865	0.562	0.570
SW-SMOTE $r=0.1$	0.101	0.558	0.587	0.903	0.879	0.648	0.646
SW-SMOTE $r=0.5$	0.042	0.468	0.402	0.862	0.852	0.386	0.387

[†]Cross-domain AUROC and AUPRC are

near-chance ($|\delta| > 0.94$ vs. Within; $\omega_{tr} = 0$, Eq. (8)) and are omitted; the F2(Cross) column is retained for completeness only.

$|\delta_{AUPRC}| = 0.03$), while Cross-domain collapses to near-chance levels ($|\delta| > 0.83$ vs. Within for F2; $|\delta| > 0.94$ for AUROC and AUPRC). The best strategy also reverses across modes: RUS $r=0.1$ ranks first in Cross-domain, whereas SW-SMOTE $r=0.1$ ranks first in Within and Mixed (Fig. 6). The Spearman rank correlation of strategy rankings between Cross and Within modes ranges from $\rho_{cross,within} = -0.74$ to -0.89 across metrics, confirming systematic reversal. SW-SMOTE $r=0.1$ with within-domain training achieves the highest threshold-independent performance (AUROC = 0.903, AUPRC = 0.648; within-domain F2 = 0.558); Mixed-mode SW-SMOTE $r=0.1$ attains a slightly higher F2 (0.587) at a more recall-heavy operating point but a lower AUROC (0.879).

D. Robustness Validation

Ranking stability (σ_{rank}) decreases monotonically with the number of seeds, reaching 0 (F2, AUPRC) or 0.147 (AUROC) by $k = 11$ of 12 seeds (Fig. 7), confirming sufficient seed count.

VI. Discussion

A. Answer to RQ1: Relative Contribution of Each Factor

The Sobol decomposition reveals a concentrated variance structure: M , R , and $R \times M$ jointly explain $> 95\%$ of systematic variance (Eq. (19)), while D and G together contribute $< 5\%$.

To understand why training mode dominates, recall the subject overlap ratio ω_{tr} (Eq. (8)). By construction, $\omega_{tr} = 0$ for $M = \text{Cross}$ (training excludes evaluation subjects), while $\omega_{tr} = 1$ for $M \in \{\text{Within}, \text{Mixed}\}$ (evaluation subjects are included). This binary switch between $\omega = 0$ and $\omega = 1$ determines whether target-subject patterns are available during training, which in turn governs the performance bottleneck and the effectiveness of rebalancing (Section VI-B).

In contrast, distance metric and domain membership are negligible ($S_{TD} < 0.015$, $S_{TG} \leq 0.031$; $BF_{01} = 71\text{--}767$). Three fundamentally different metrics—MMD (distributional), DTW (temporal), and Wasserstein (geometric)—produce equivalent downstream performance despite generating different subject rankings (pairwise Spearman ρ :

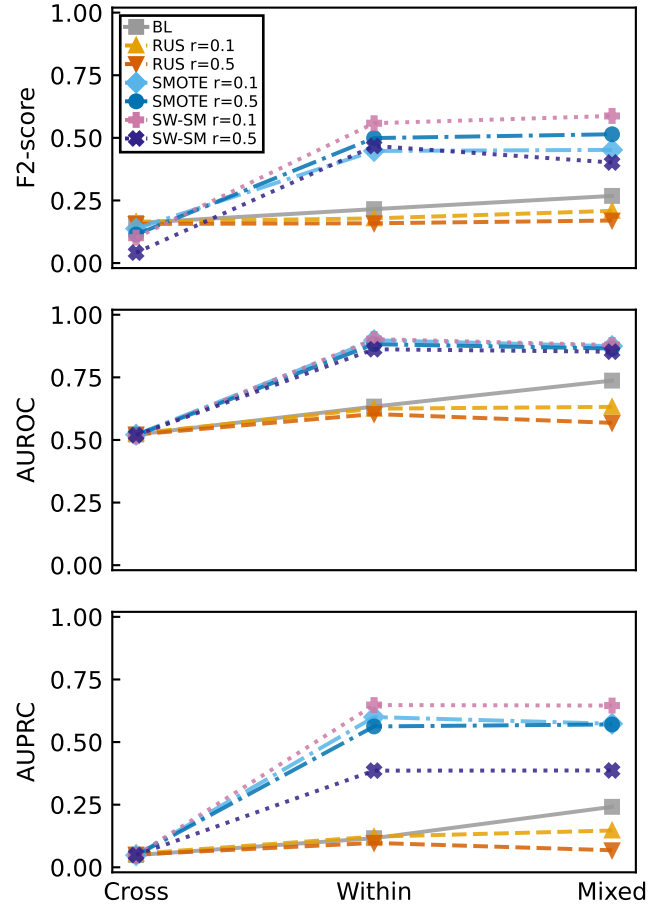


Fig. 6. $R \times M$ interaction: rebalancing strategy performance conditioned on training mode (rows: F2-score, AUROC, AUPRC; columns: training modes; lines: rebalancing strategies, see legend). In Cross-domain, all strategies collapse to near chance; in Within/Mixed, SMOTE-based strategies improve sharply while RUS variants remain low, confirming a strong ranking reversal across modes.

0.12–0.76; DTW vs. MMD: $\rho = 0.12$, $p = 0.29$; up to 39.1% of subjects switch domain partition).

This metric invariance stems from three factors: the bicycle model (Eqs. (10)–(11)) compresses the effective dimensionality of the unified 145-dimensional feature space (Section IV-C2) to $p_{eff} = 49$ (95% PCA variance, with the first principal component alone explaining 20.4%), low per-feature inter-subject identifiability (mean ICC = 0.111

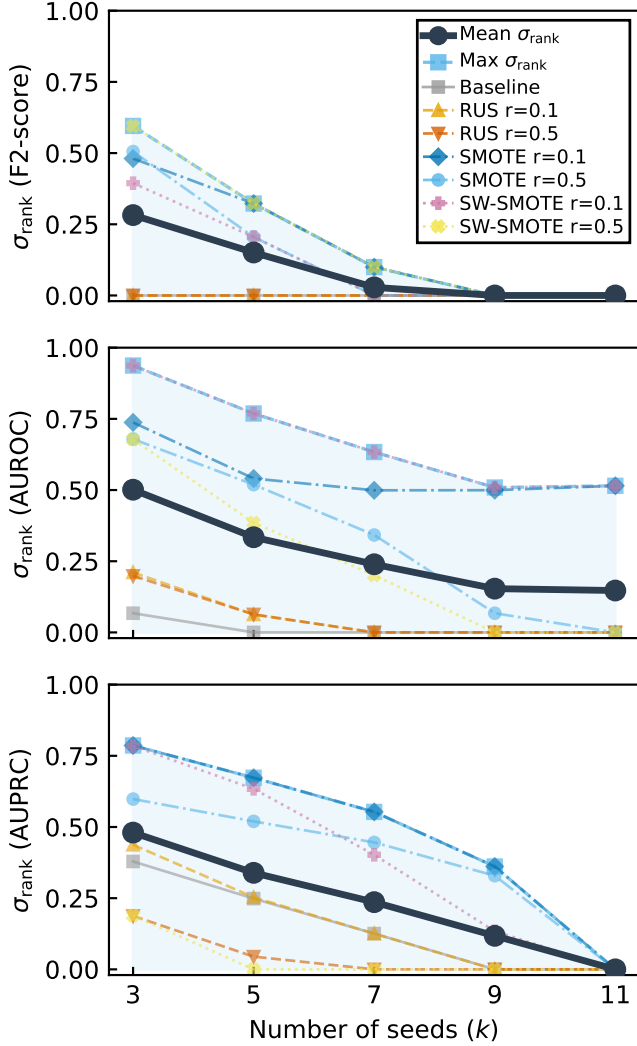


Fig. 7. Ranking stability (σ_{rank}) as a function of seed subset size k (rows: F2-score, AUROC, AUPRC). Thick line: mean σ_{rank} across the seven strategies; dashed line with shaded band: maximum σ_{rank} . Thin lines show per-strategy convergence (see legend). Monotonic convergence confirms that $n = 12$ seeds is sufficient.

across the same 145 features; see Section IV-D) makes partition relabelling inconsequential despite up to 39.1% of subjects switching domain (34 of 87 subjects), and the dominant rebalancing factor absorbs residual grouping differences ($S_{TR}/S_{TD} = 26\text{--}31\times$; $BF_{01} > 70$).

B. Answer to RQ2: Interaction and Optimal Factor Combination

The $R \times M$ interaction ($S_{R \times M} = 0.212$ for F2, concentrated at 87–96% of each dominant factor’s interaction variance) confirms that the contributing factors do not act independently. This interaction manifests as a mode-dependent strategy reversal: the identity of the best rebalancing strategy changes across training modes (Table IV; Fig. 6). A practitioner who selects a rebalancing strategy based on cross-domain results would deploy the worst strategy for within-domain operation, and vice versa.

The mechanistic origin is a mode-dependent performance bottleneck. The mode-dependent rebalancing gain, averaged over D and G , is:

$$\Delta Y_R^{(M)} = \mathbb{E}_{D,G}[Y_{D,G,M,R} - Y_{D,G,M,BL}] \quad (20)$$

For the best strategy (SW-SMOTE $r=0.1$), the empirical values are:

$$\Delta Y_{\text{SW-SM}}^{(M)} = \begin{cases} +0.343 (+159\%) & M = \text{Within } (\omega = 1) \\ +0.319 (+119\%) & M = \text{Mixed } (\omega = 1) \\ -0.059 (-37\%) & M = \text{Cross } (\omega = 0) \end{cases} \quad (21)$$

When $\omega_{\text{tr}} = 1$, the performance bottleneck is class imbalance ($\rho \ll 1$), and oversampling is effective. When $\omega_{\text{tr}} = 0$, the bottleneck shifts to distributional mismatch ($P_{\text{pool}}(\mathbf{x}) \neq P_{\text{test}}(\mathbf{x})$), and all strategies collapse to near-chance levels ($F2 = 0.04\text{--}0.17$). This sign reversal of $\Delta Y_R^{(M)}$ (Eq. (21)) is the mechanistic origin of the $R \times M$ interaction. The $G \times M$ interaction is negligible ($S_{G \times M} = 0.006\text{--}0.009$), an order of magnitude below $S_{R \times M}$. The optimal factor combination is therefore SW-SMOTE $r=0.1$ with $\omega_{\text{tr}} = 1$; within-domain training maximizes threshold-independent discrimination (AUROC = 0.903, AUPRC = 0.648; within-domain F2 = 0.558), while Mixed mode trades 0.024 AUROC for a higher recall-leaning F2 (0.587).

C. Practical Implications

The Sobol total-order indices establish a strict factor priority ordering for practitioners deploying vehicle-dynamics-based DDD systems:

$$S_{TM} (0.48\text{--}0.66) > S_{TR} (0.39\text{--}0.46) \gg S_{TG} (\leq 0.031) \approx S_{TD} (< 0.015) \quad (22)$$

This ordering, combined with the subject overlap ratio ω_{tr} (Eq. (8)), yields the following decision rules:

- 1) Training mode (highest priority): Use within-domain or mixed training whenever target-subject data is available ($\omega_{\text{tr}} = 1$).
- 2) Class rebalancing (second priority): The optimal rebalancing strategy is mode-dependent (Eq. (21)). When $\omega_{\text{tr}} = 1$, apply SW-SMOTE at $r = 0.1$:

$$R^*(\omega_{\text{tr}}) = \begin{cases} \text{SW-SMOTE, } r=0.1 & \omega_{\text{tr}}=1 \\ \text{any (none helps)} & \omega_{\text{tr}}=0 \end{cases} \quad (23)$$

where $\Delta Y = +159\%$ (F2) for $\omega_{\text{tr}} = 1$ and $\Delta Y \leq 0$ for $\omega_{\text{tr}} = 0$.

- 3) Distance metric and domain membership (negligible): Use any convenient metric for domain grouping. The choice has no measurable impact on classification performance ($BF_{01} > 70$).
- 4) Bridging the cross-domain gap ($\omega_{\text{tr}} = 0$). Since no tested rebalancing strategy improves cross-domain performance (Eq. (21)), closing this gap requires reducing the distributional mismatch $P_{\text{pool}}(\mathbf{x}) \neq$

TABLE V

Comparison with Prior DDD Studies. “Factors”: number of design factors systematically varied; “Interaction”: whether inter-factor interactions were quantified. Most prior studies report accuracy and/or F1 instead of F2/AUROC/AUPRC; the “Other reported” column lists the headline metric(s) given in the source. NR = not reported.

Study	Year	Classif.	Modal.	Subj.	Fact.	Inter.	F2	AUROC	AUPRC	Other reported
Arefnezhad et al. [10]	2019	SVM	Vehicle	39	0	No	NR	0.97 [†]	NR	Acc=98.1%
Zhu et al. [15]	2025	RF	Vehicle	25	0	No	NR	NR	NR	Acc/P/R = 93.7/93.7/93.9%
Baccour et al. [11]	2022	SVM	Vehicle	70	0	No	NR	NR	NR	Bal. Acc=87.1%
Sun et al. [12]	2021	Fisher	Vehicle	35	0	No	NR	NR	NR	Acc=85.3%, F1=79.6%
Yadawadkar et al. [18]	2018	NB	Vehicle	—	1 (imbalance)	No	NR	NR	NR	F1=94% (NB+SMOTE)
Schwarz et al. [19]	2019	RF	Vehicle	20	0	No	NR	NR	NR	5-fold Acc=0.94–0.95
He et al. [40]	2024	GARMFCN	EEG+EOG	23	1 (imbalance)	No	NR	0.87–0.96 [§]	NR	Acc=84.7%, w-F1=0.85
Kim et al. [38]	2025	PDMix	EEG	27	1 (domain)	No	NR	0.738 [‡]	NR	F1=70.3%
Ours	2026	RF	Vehicle	87	4 (R, M, D, G)	$R \times M$ Sobol	0.558	0.903	0.648	—

[†] Evaluation protocol not specified in source; based on the reported sample counts (35,079 test samples drawn from 39 drivers’ 20 h of pooled data), the most plausible default is a window-level random split (effectively within-subject), which is not directly comparable to our per-subject 70/15/15 chronological split.

[‡] Cross-subject evaluation (PDMix is a domain-generalization method evaluated on subject-disjoint splits); directly comparable to our cross-domain ($\omega_{tr}=0$) results, not the within-domain AUROC = 0.903 reported in the same row.

[§] Per-class one-vs-rest AUC for a 3-class (awake/tired/drowsy) classifier on EEG+EOG; not directly comparable to our binary AUROC. The reported per-class values are 0.87 (awake), 0.88 (tired), 0.96 (drowsy).

$P_{\text{test}}(\mathbf{x})$ directly. Promising directions include unsupervised domain adaptation [6], [7], few-shot personalization [38], and continual learning. Each can be introduced as an additional factor level within the same Sobol decomposition pipeline, yielding directly comparable variance shares.

D. Comparison with Prior Work

Our finding that class imbalance handling is the dominant design factor aligns with Taylor et al. [39], who showed that data-related hyperparameters dominate architectural choices in deep learning. However, our study is the first to demonstrate this in the DDD context and to reveal the interaction with training mode that causes mode-dependent strategy ranking change. Prior DDD studies that compare rebalancing methods under a single training mode [18], [40] risk selecting a suboptimal strategy for deployment.

The best within-domain performance (F2=0.558, AUROC=0.903) is competitive with recent vehicle-dynamics-based methods. Table V positions this work relative to prior DDD studies. Direct performance comparison is limited because prior studies use different datasets, modalities, and evaluation protocols; the primary contribution here is the factorial decomposition framework, not absolute classification performance.

The distance metric irrelevance contrasts with the implicit assumption in domain adaptation literature [6], [38] that domain definition substantially impacts downstream performance; the three-layer mechanism identified in Section VI-A explains why this assumption does not hold for vehicle-dynamics features.

E. Limitations

- 1) Single classifier family. The factorial experiment uses RF [28]. The Sobol–Hoeffding framework is

classifier-agnostic and can be extended to other learners (e.g., LSTM, SVM) without methodological modification.

- 2) Deterministic data split. The train/test partition is deterministic; random seeds vary only model initialisation and resampling.
- 3) Simulator data. All data come from a laboratory driving simulator. On-road conditions introduce additional variability that may alter the absolute variance shares; validation on naturalistic datasets (e.g., SHRP 2) is needed.
- 4) Sample size per cell. With $n = 12$, only large effects ($|\delta| > 0.53$) are detectable after Bonferroni correction.
- 5) Feature selection scope. Only the top $k = 10$ features from the unified $p = 145$ feature pool are used in the classifier; sensitivity to k and to alternative feature subsets remains to be tested.
- 6) Binary labeling threshold. $\text{KSS} \leq 5$ vs. $\text{KSS} \geq 8$ excludes intermediate drowsiness (KSS 6–7), which may limit applicability to graded drowsiness detection scenarios.
- 7) Calibration–threshold coupling. When $M \in \{\text{Within, Mixed}\}$, the calibration pool $\mathcal{S}_{tr}$ includes $\mathcal{G}_{eval}$, so $\mathcal{G}_{eval}$ ’s validation windows serve both as part of the 5-fold calibration data and as the support for the F2-optimal threshold $\hat{\tau}$; $\hat{\tau}$ is therefore mildly optimistic on the validation set for these two modes. When $M = \text{Cross}$, calibration and threshold-optimization pools are drawn from disjoint subject sets, so no coupling arises. Final test metrics are unaffected in all modes because the test split is held out from HPO, calibration, and threshold selection.

VII. Conclusion

This study quantifies the relative importance of four design factors in vehicle-dynamics-based DDD through a factorial experiment with Sobol–Hoeffding variance decomposition over 1,512 evaluated random forest models from 87 drivers. Training mode ($S_{TM} = 0.48\text{--}0.66$) and rebalancing ($S_{TR} = 0.39\text{--}0.46$), including their interaction ($S_{R \times M} = 0.21$), account for $> 95\%$ of systematic variance, whereas distance metric ($S_{TD} < 0.015$; $BF_{01} = 71\text{--}767$) and domain membership ($S_{TG} \leq 0.031$) are negligible. The $R \times M$ interaction reverses the optimal strategy across modes ($\rho_{\text{cross, within}} = -0.74$ to -0.89): when evaluation subjects overlap with training data ($\omega_{\text{tr}} = 1$), class imbalance is the bottleneck and oversampling is effective; when they do not ($\omega_{\text{tr}} = 0$), distributional mismatch dominates and no tested rebalancing strategy helps. SW-SMOTE $r=0.1$ with within-domain training ($\omega_{\text{tr}} = 1$) achieves the highest threshold-independent performance (AUROC = 0.903, AUPRC = 0.648; within-domain F2 = 0.558). Practitioners should prioritize training mode and class rebalancing (Eq. (22)); domain configuration requires no optimization.

Future work includes validation with other classifiers (LSTM, SVM), on naturalistic driving data (SHRP 2), and incorporation of domain adaptation and few-shot methods as additional factor levels.

Data and Code Availability

The driving simulator dataset is publicly available at Harvard Dataverse (DOI: 10.7910/DVN/HMZ5RG) [25]. Analysis code and trained models will be released upon publication.

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References

- [1] World Health Organization, “Global status report on road safety 2018,” World Health Org., Geneva, Switzerland, Tech. Rep., 2018.
- [2] B. Fu, F. Boutros, C.-T. Lin, and N. Damer, “A survey on drowsiness detection—modern applications and methods,” *IEEE Trans. Intell. Veh.*, vol. 9, no. 11, pp. 7279–7303, 2024.
- [3] T. Fonseca and S. Ferreira, “Drowsiness detection in drivers: A systematic review of deep learning-based models,” *Appl. Sci.*, vol. 15, no. 16, p. 9018, 2025.
- [4] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, “SMOTE: Synthetic minority over-sampling technique,” *J. Artif. Intell. Res.*, vol. 16, pp. 321–357, 2002.
- [5] H. He and E. A. Garcia, “Learning from imbalanced data,” *IEEE Trans. Knowl. Data Eng.*, vol. 21, no. 9, pp. 1263–1284, Sep. 2009.
- [6] S. J. Pan and Q. Yang, “A survey on transfer learning,” *IEEE Trans. Knowl. Data Eng.*, vol. 22, no. 10, pp. 1345–1359, Oct. 2010.
- [7] K. Weiss, T. M. Khoshgoftaar, and D. Wang, “A survey of transfer learning,” *J. Big Data*, vol. 3, no. 1, p. 9, 2016.
- [8] G. Douzas, F. Bacao, and F. Last, “Improving imbalanced learning through a heuristic oversampling method based on k-means and SMOTE,” *Inf. Sci.*, vol. 465, pp. 1–20, 2018.
- [9] P. Bota, R. Silva, C. Carreiras, A. Fred, and H. P. da Silva, “A comparative study in class imbalance mitigation when working with physiological signals,” *Frontiers in Digital Health*, vol. 6, p. 1377165, 2024.
- [10] S. Arefnezhad, S. Samiee, A. Eichberger, and A. Nahvi, “Driver drowsiness detection based on steering wheel data applying adaptive neuro-fuzzy feature selection,” *Sensors*, vol. 19, no. 4, p. 943, 2019.
- [11] M. H. Baccour, F. Driewer, T. Schäck, and E. Kasneci, “Comparative analysis of vehicle-based and driver-based features for driver drowsiness monitoring by support vector machines,” *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 12, pp. 25 066–25 079, 2022.
- [12] Y. Sun, C. Wu, H. Zhang, Y. Zhang, and S. Li, “Extraction of optimal measurements for drowsy driving detection considering driver fingerprinting differences,” *J. Adv. Transp.*, vol. 2021, p. 5546127, 2021.
- [13] C. Zhao et al., “Driver drowsiness detection using continuous wavelet transform,” in *Proc. IEEE Int. Conf. Signal Process.*, 2012, pp. 1–4.
- [14] X. Wang, R. Xu, S. Zhang, Y. Zhuang, and Y. Wang, “Driver distraction detection based on vehicle dynamics using naturalistic driving data,” *Transp. Res. Part C, Emerg. Technol.*, vol. 136, p. 103561, 2022.
- [15] J. Zhu, Q. Liu, Z. Zhao, Y. Lu, Z. Yao, and Y. Li, “MADFM: Multi-layer adaptive driver fatigue monitoring model based on steering wheel signals,” *IEEE Trans. Intell. Transp. Syst.*, 2025.
- [16] M. S. Al-Quraishi et al., “Technologies for detecting and monitoring drivers’ states: A systematic review,” *Heliyon*, vol. 10, no. 21, p. e38592, 2024.
- [17] E. Perkins, C. Sitaula, M. Burke, and F. Dehzangi, “Challenges of driver drowsiness prediction: The remaining steps to implementation,” *IEEE Trans. Intell. Veh.*, vol. 8, no. 12, pp. 4704–4716, 2022.
- [18] S. Yadawadkar, B. Mayer, and S. Lokegaonkar, “Identifying distracted and drowsy drivers using naturalistic driving data,” in *Proc. IEEE Int. Conf. Big Data*, 2018, pp. 2889–2898.
- [19] C. Schwarz, J. Gaspar, and T. Miller, “The detection of drowsiness using a driver monitoring system,” *Traffic Injury Prev.*, vol. 20, no. sup2, pp. S157–S161, 2019.
- [20] S. S. Aravindh, G. M. Nagamani, C. K. Kumar, and A. Lasisi, “Dynamic cross-domain transfer learning for driver fatigue monitoring,” *Sci. Rep.*, vol. 15, p. 92701, 2025.
- [21] M. Maritsch, K. Koch, H. Thomsen, and N. Kühl, “Driver state prediction from vehicle signals: An evaluation of segmentation approaches,” in *Proc. IEEE Int. Conf. Intell. Transp. Syst. (ITSC)*, 2022, pp. 1–6.
- [22] A. Gretton, K. M. Borgwardt, M. J. Rasch, B. Schölkopf, and A. Smola, “A kernel two-sample test,” *J. Mach. Learn. Res.*, vol. 13, no. 1, pp. 723–773, 2012.
- [23] D. J. Berndt and J. Clifford, “Using dynamic time warping to find patterns in time series,” in *Proc. AAAI Workshop Knowl. Discovery Databases*, Seattle, WA, USA, 1994, pp. 359–370.
- [24] C. Villani, *Optimal Transport: Old and New*. Berlin, Germany: Springer, 2009.
- [25] A. Aygun et al., “Multi-modal data acquisition platform for behavioral evaluation,” Harvard Dataverse, 2024, DOI: 10.7910/DVN/HMZ5RG.
- [26] R. Rajamani, *Vehicle Dynamics and Control*, 2nd ed. New York, NY, USA: Springer, 2012.
- [27] T. Åkerstedt and M. Gillberg, “Subjective and objective sleepiness in the active individual,” *Int. J. Neurosci.*, vol. 52, no. 1–2, pp. 29–37, 1990.
- [28] L. Breiman, “Random forests,” *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, 2001.
- [29] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, “Optuna: A next-generation hyperparameter optimization framework,” in *Proc. ACM SIGKDD Int. Conf. Knowl. Discov. Data Mining*, 2019, pp. 2623–2631.
- [30] J. C. Platt, “Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods,” in *Advances in Large Margin Classifiers*. MIT Press, 1999, pp. 61–74.
- [31] T. Saito and M. Rehmsmeier, “The precision–recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets,” *PLOS ONE*, vol. 10, no. 3, p. e0118432, Mar. 2015.

- [32] A. Saltelli, M. Ratto, T. Andres, F. Campolongo, J. Cariboni, D. Gatelli, M. Saisana, and S. Tarantola, *Global Sensitivity Analysis: The Primer*. Chichester, UK: Wiley, 2008.
- [33] I. M. Sobol', "Sensitivity estimates for nonlinear mathematical models," *Math. Modelling Comput. Exp.*, vol. 1, no. 4, pp. 407–414, 1993.
- [34] N. Cliff, "Dominance statistics: Ordinal analyses to answer ordinal questions," *Psychol. Bull.*, vol. 114, no. 3, pp. 494–509, 1993.
- [35] E.-J. Wagenmakers, "A practical solution to the pervasive problems of p values," *Psychon. Bull. Rev.*, vol. 14, no. 5, pp. 779–804, 2007.
- [36] H. Jeffreys, *Theory of Probability*, 3rd ed. Oxford, UK: Oxford Univ. Press, 1961.
- [37] P. E. Shrout and J. L. Fleiss, "Intraclass correlations: Uses in assessing rater reliability," *Psychol. Bull.*, vol. 86, no. 2, pp. 420–428, 1979.
- [38] D. Y. Kim, D. K. Han, and J. H. Jeong, "Calibration-free driver drowsiness classification with prototype-based multi-domain mixup," *IEEE Trans. Intell. Transp. Syst.*, 2025.
- [39] R. Taylor, V. Ojha, I. Martino, and G. Nicosia, "Sensitivity analysis for deep learning: Ranking hyper-parameter influence," in *Proc. IEEE Symp. Ser. Comput. Intell.*, 2021, pp. 1–10.
- [40] L. He, L. Zhang, Q. Sun, and X. Lin, "A generative adaptive convolutional neural network with attention mechanism for driver fatigue detection with class-imbalanced and insufficient data," *Behav. Brain Res.*, vol. 464, p. 114898, Apr. 2024.

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